HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.

GENERATIVE AI IN BIOINFORMATICS, MEDICINE AND HEALTHCARE

Chapter 6: Explainable AI

ONE LOVE. ONE FUTURE.

Content

6.1. Introduction to Explainable AI

6.2. Explainable AI in GenAI

6.3. Several interpretation methods can be applied to Gen AI

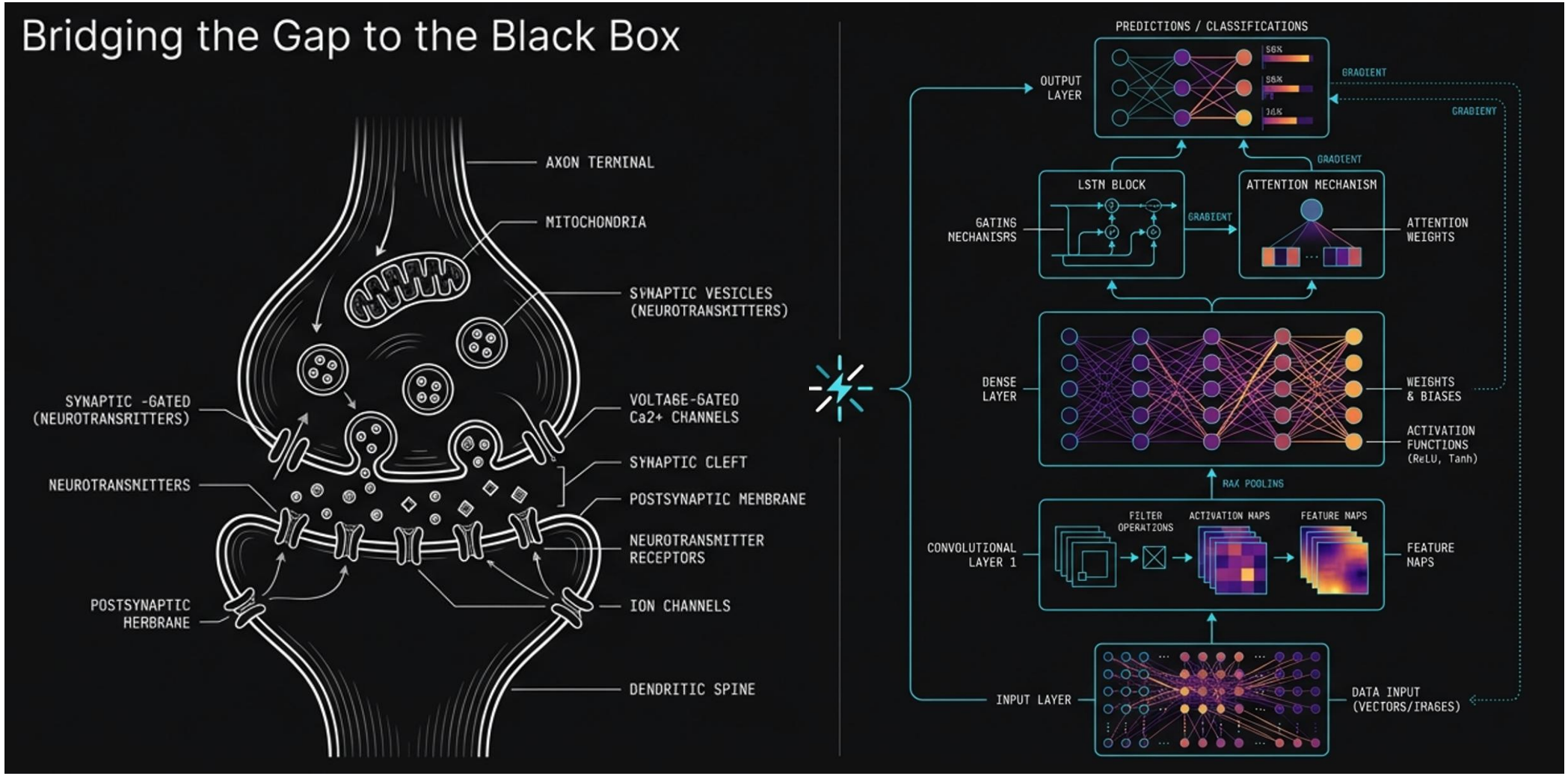
6.4. Evaluate interpretation by incorporating experimental data

6.5 Identify key biological factors in AI models

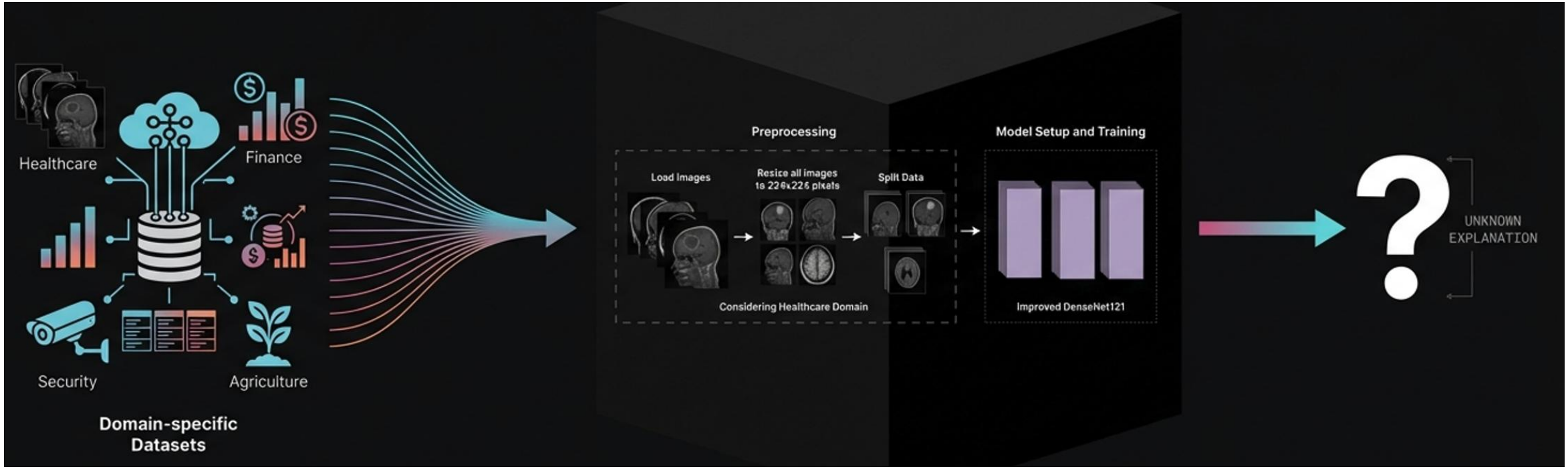
6.6. Interpretation in modern problems

6.1. Introduction to Explainable AI

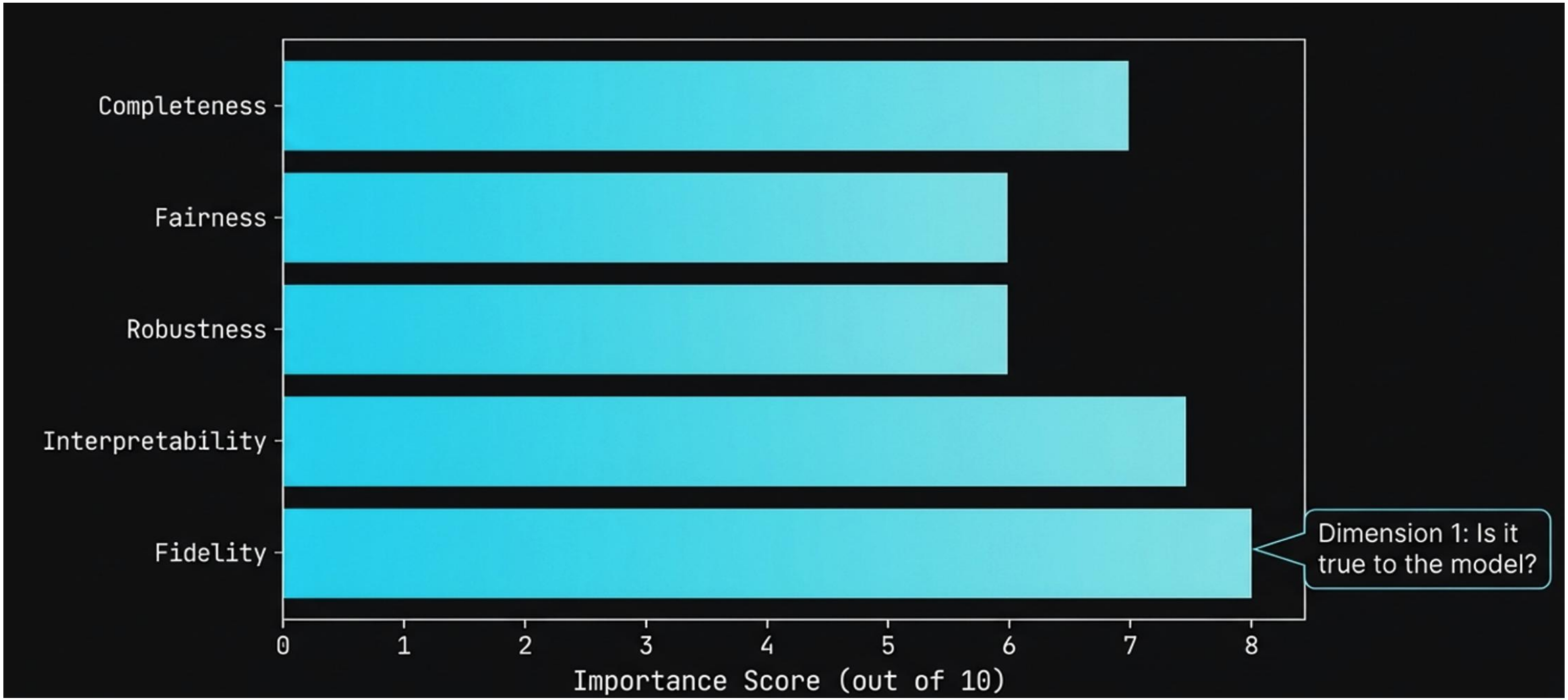
What is inside black boxes



- Deep learning models function as opaque decision-makers, as performance rises , interpretability plummets , creating a crisis of trust in high-stakes fields .

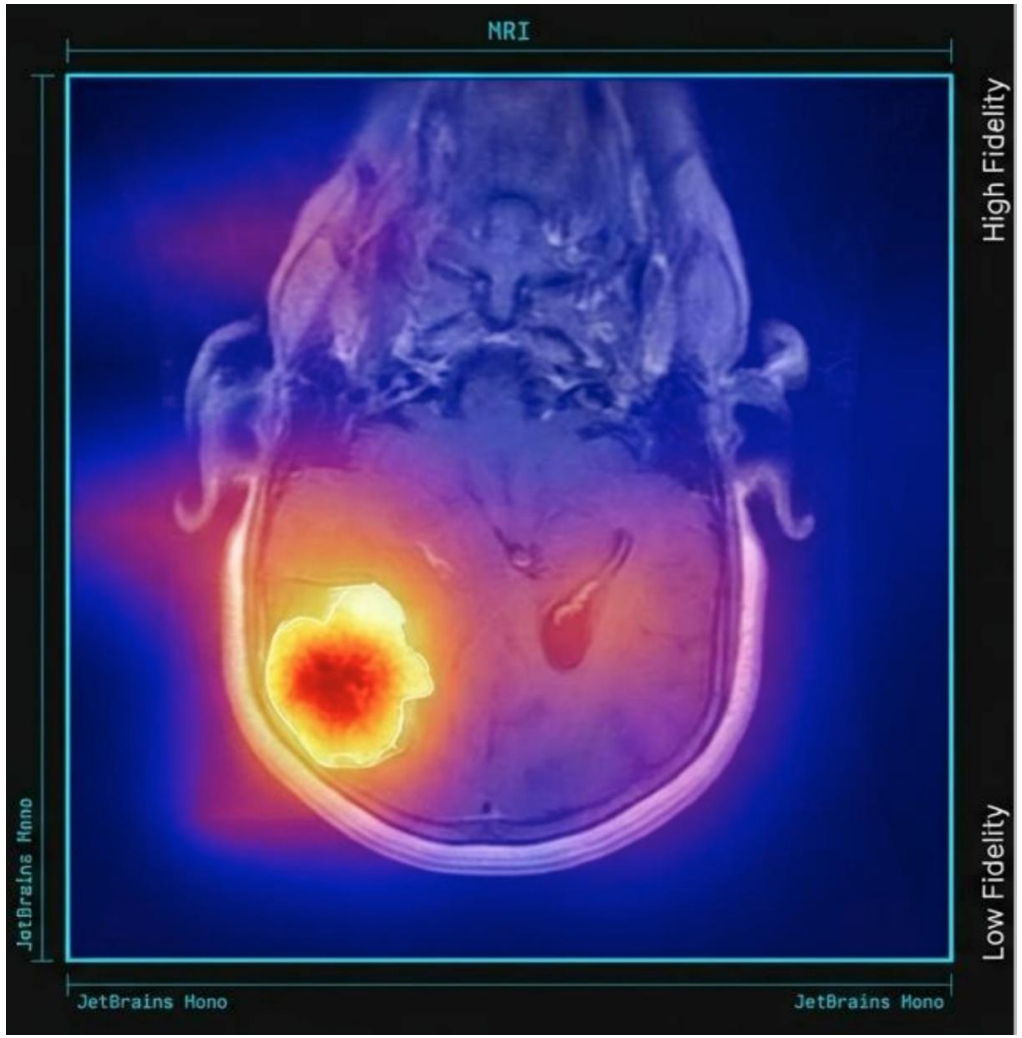


Dimensions for evaluating



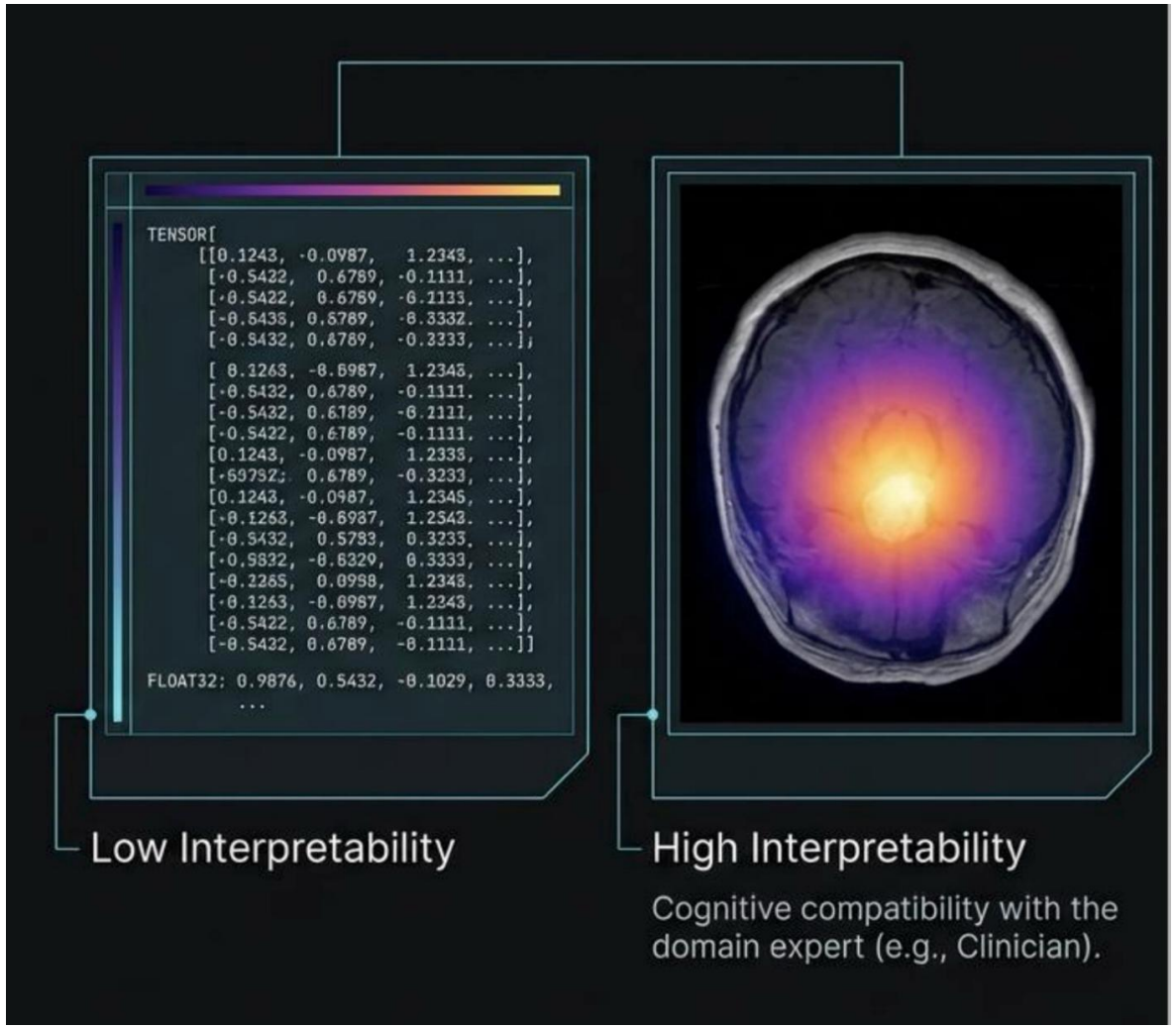
Dimension 1: Fidelity

The mirror of Truth:
How accurately an explanation, surrogate, or compressed model reflects the behavior of the original model.

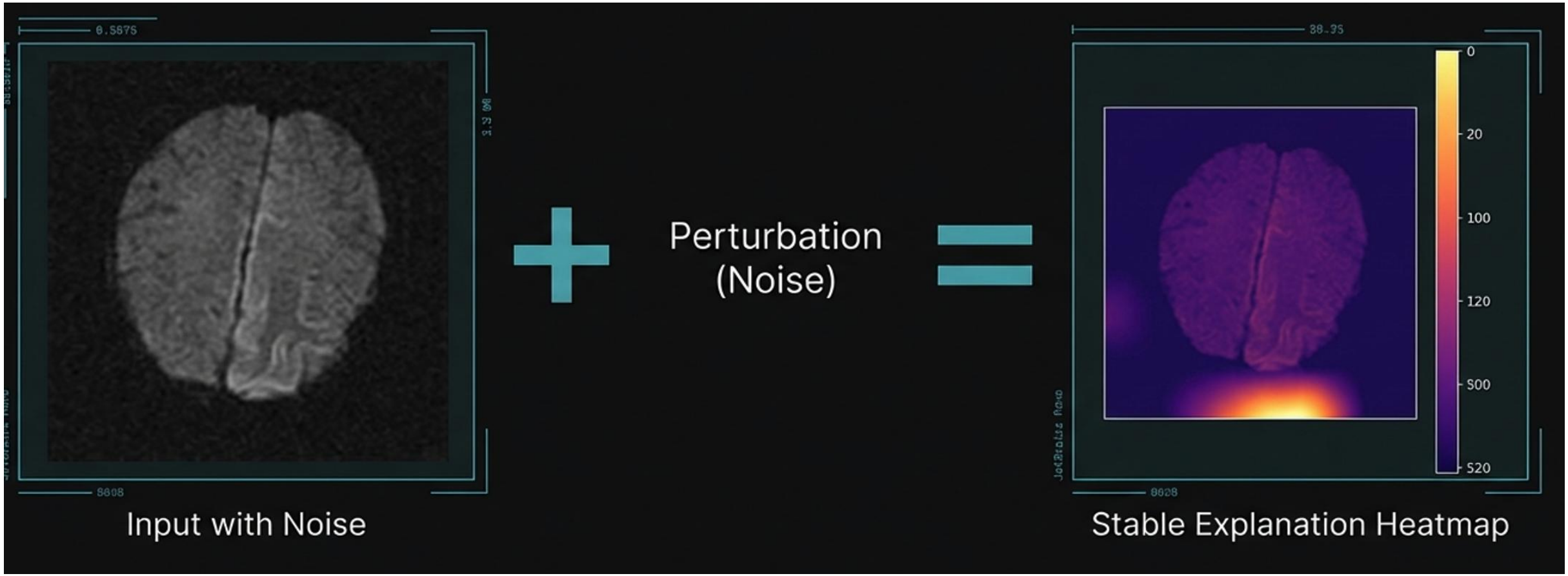


Dimension 2: Interpretibility

- Provide Human Interface
  - Can human understand?



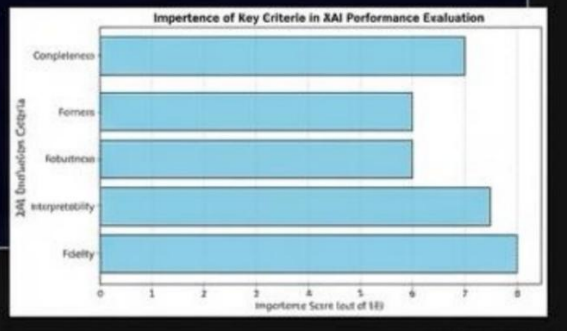
Dimension 3: Robustness



Robustness measures if the explanation holds steady when the input is slightly perturbed . If the explanation shatters under noise , it is unreliable .

Dimension 4+5: Fairness and Completness

Fairness: Neutrality across demographics.



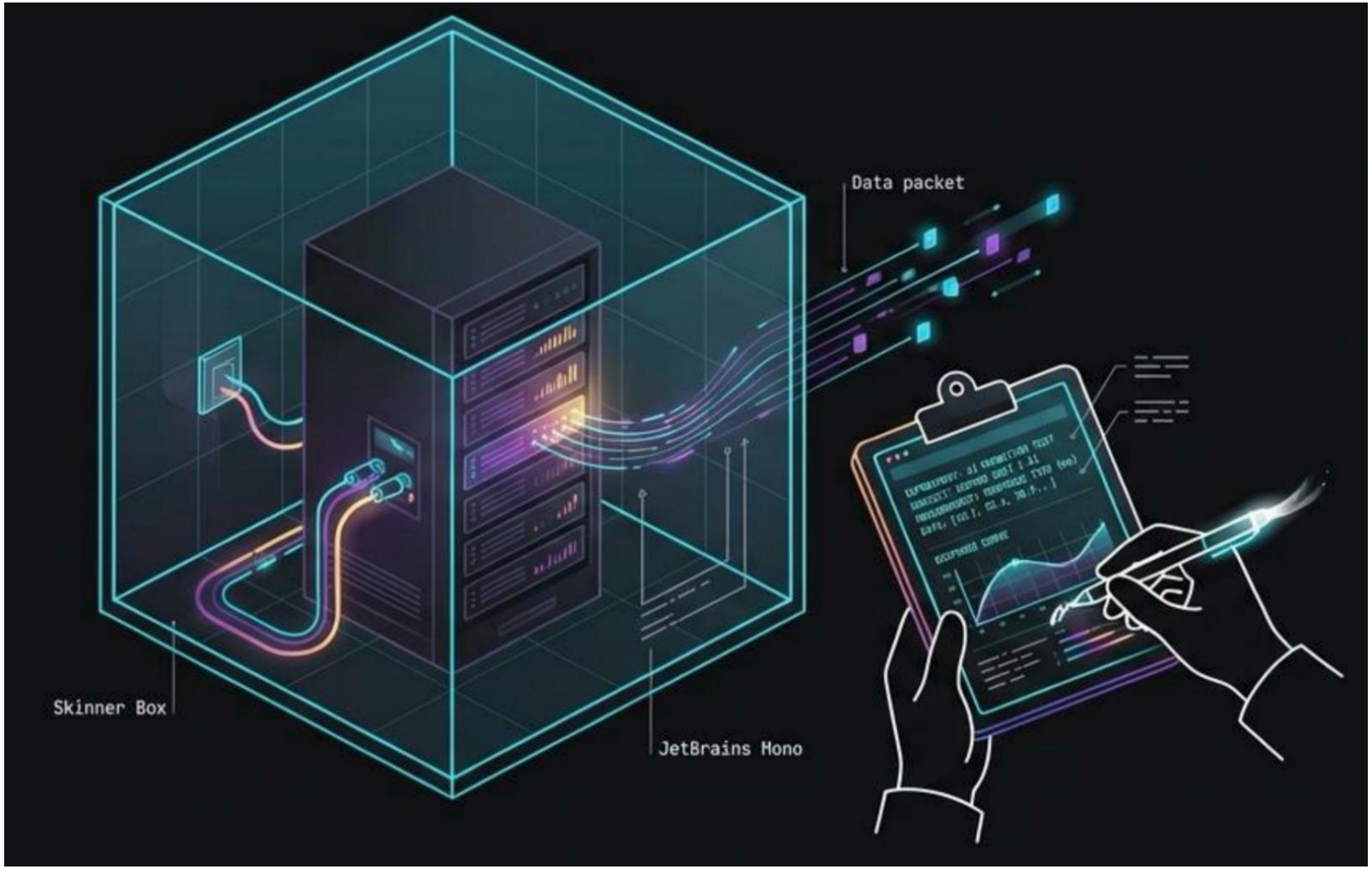
Completeness: Does removing the feature destroy the prediction?

The disagreement Problem

"This gap in utility is a wake-up call."

(ABC: Attribution-Based Confidence, BND: Baseline-Noise Difference, PGI: Prediction Gap on Importance)

From Engineer to Artificial Cognition

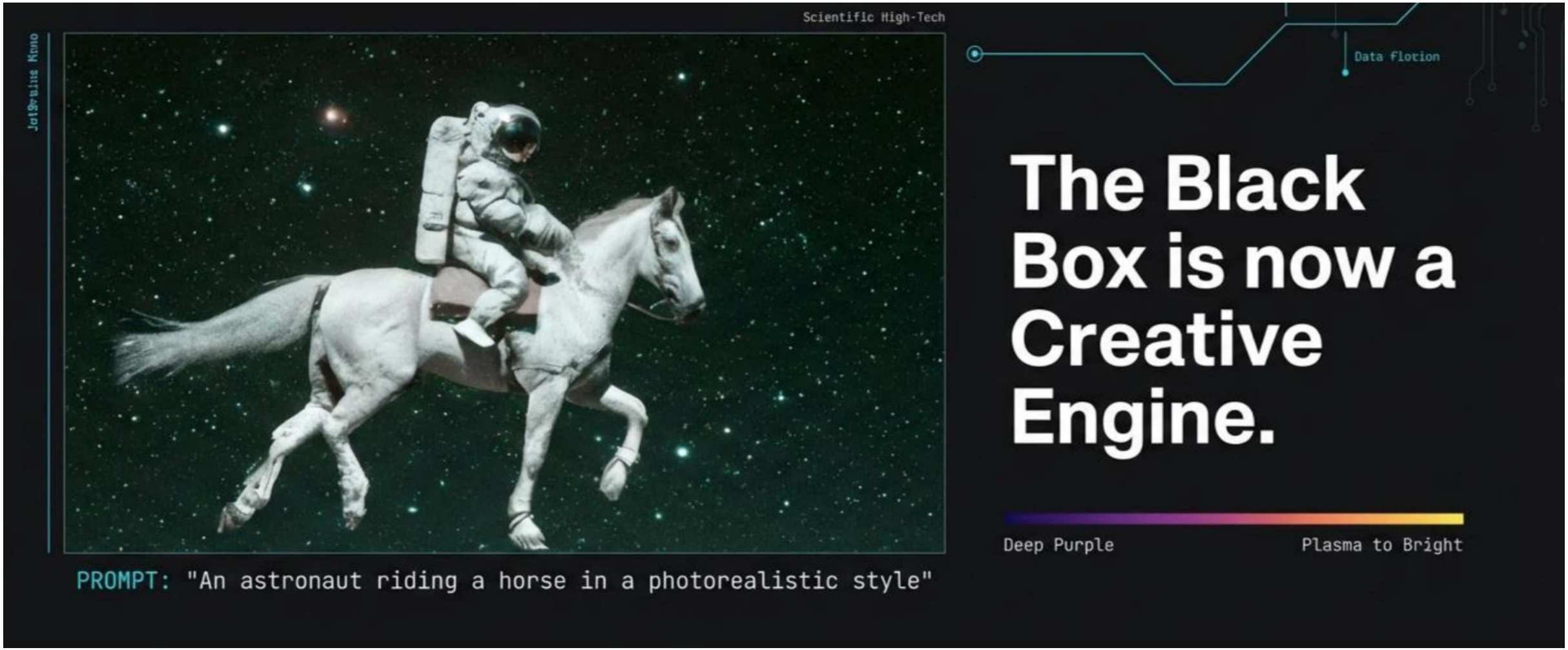


- Psychophysics for Machines
- Treating the AI as a subject in a psychology experiment
- Measuring behavior (response times) rather than weights

HUST

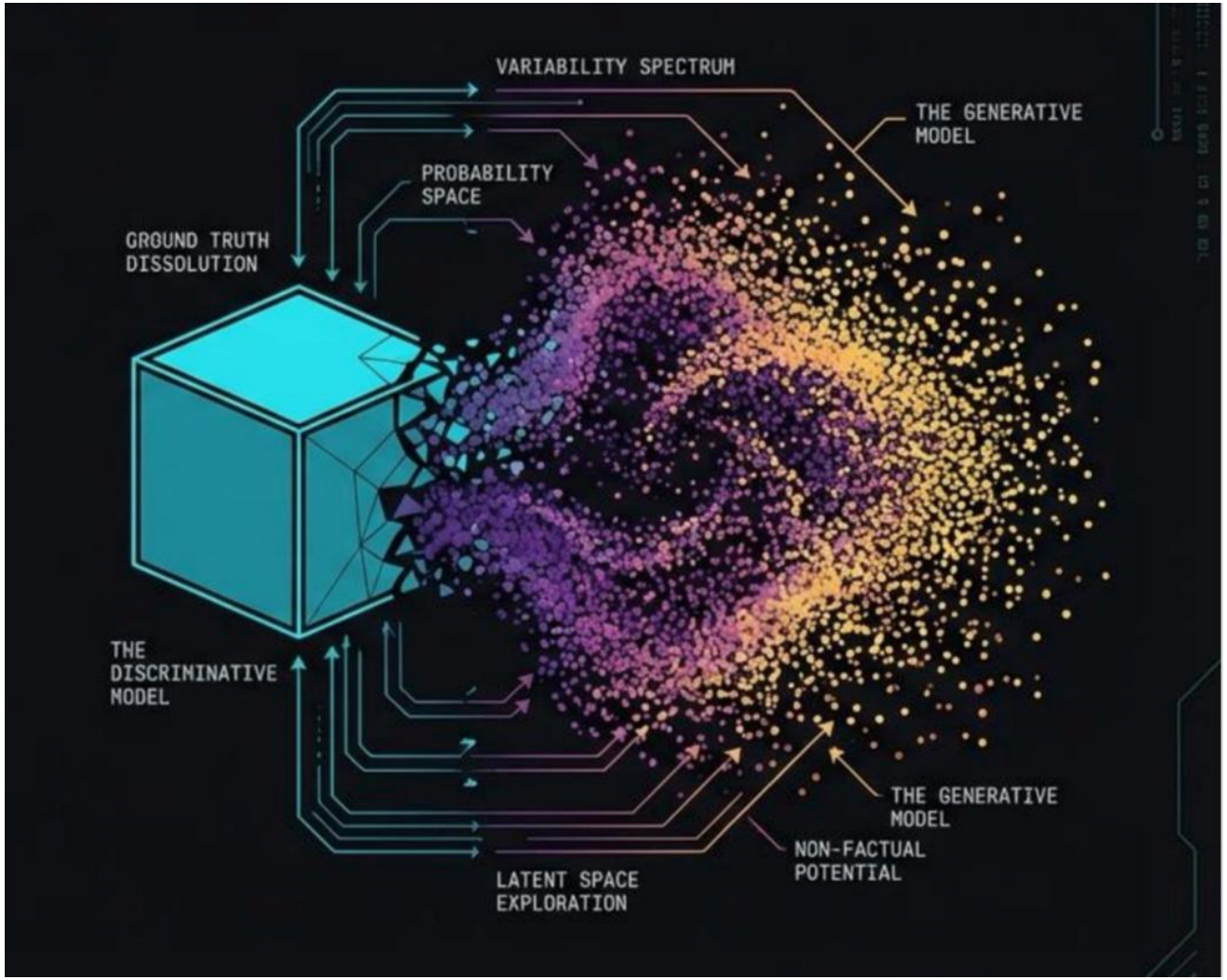
6.2. Explainable AI in GenAI

From Decision to Creation

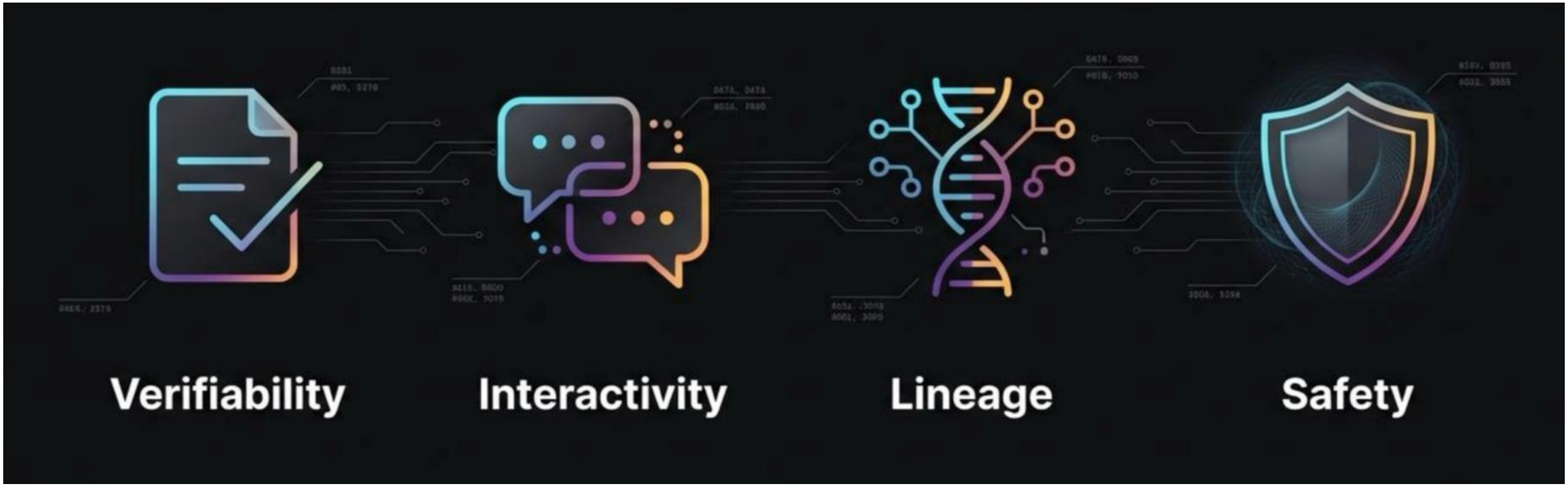


Explainable Generative AI

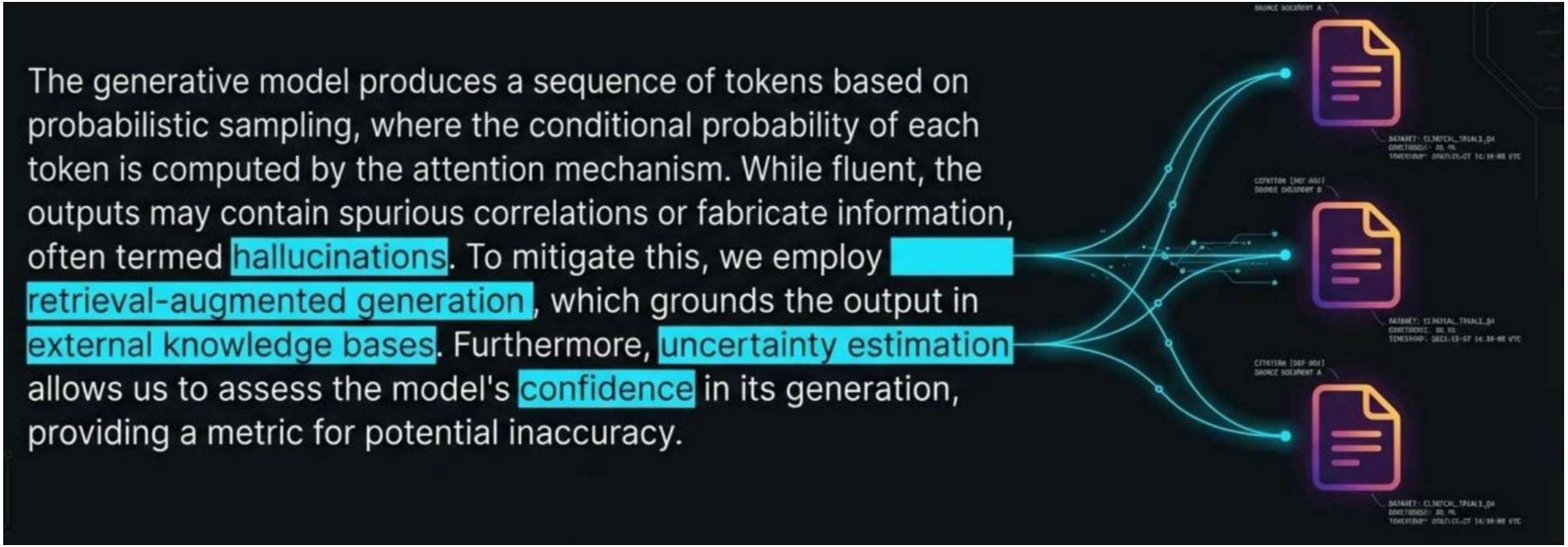
- The Challenge : Infinite output variability
- The Risk : Fluent but Nonfactual (Hallucinations)
- The Need : Prompt Engineering control



Expecting in GenAI



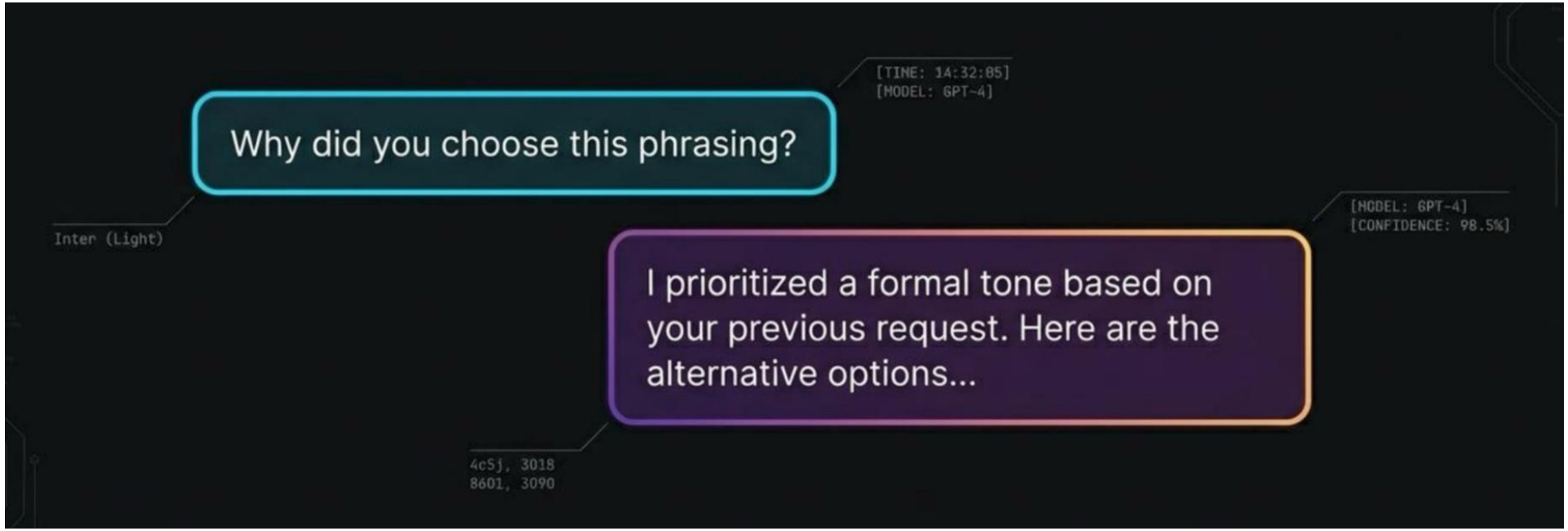
Combating Hallucination: Verfiability



XAI might the verification of generated content through
Uncertainty Estimation and RAG (Retrieval Augmented Generation).

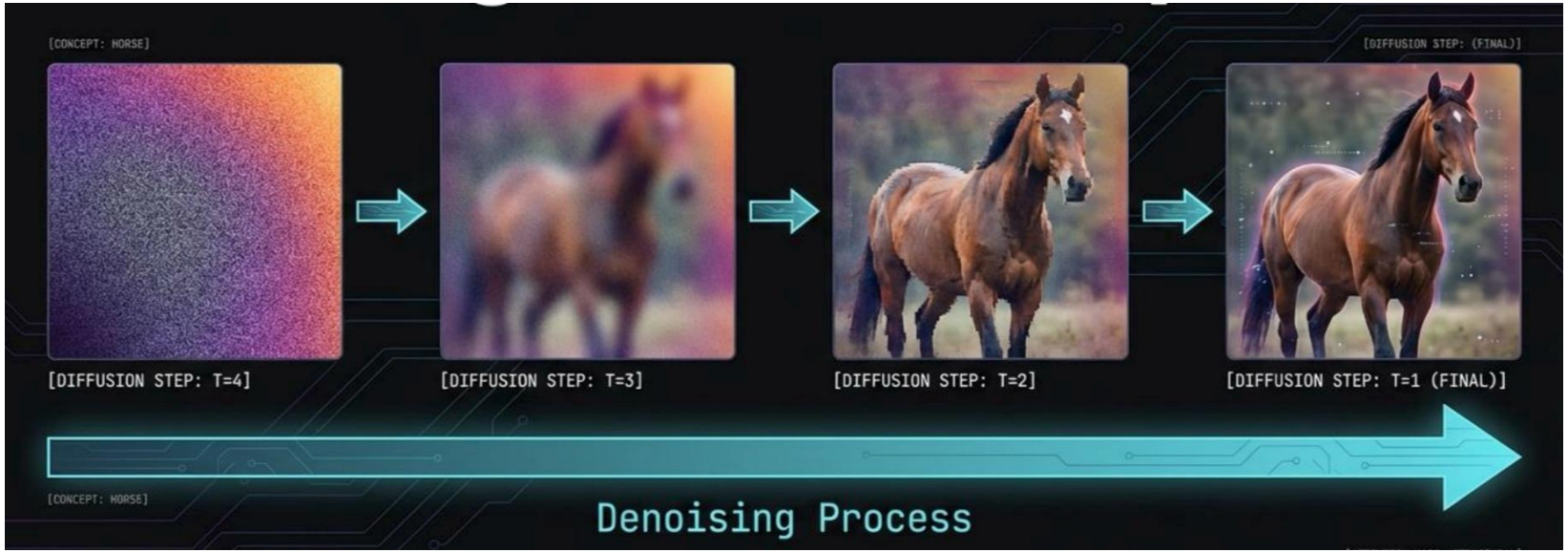
Explaination as Dialoge

- Moving from static heatmaps to iterative, conversational refinement

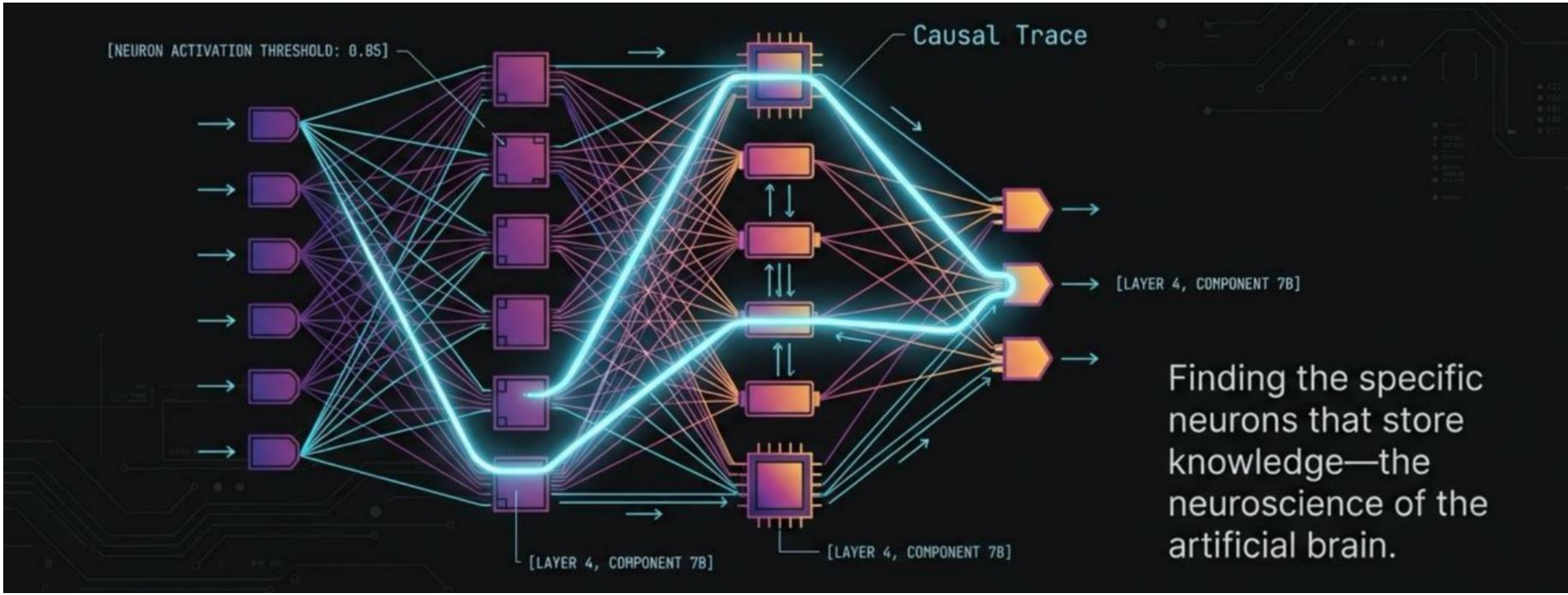


Decoding in Latent Spaces

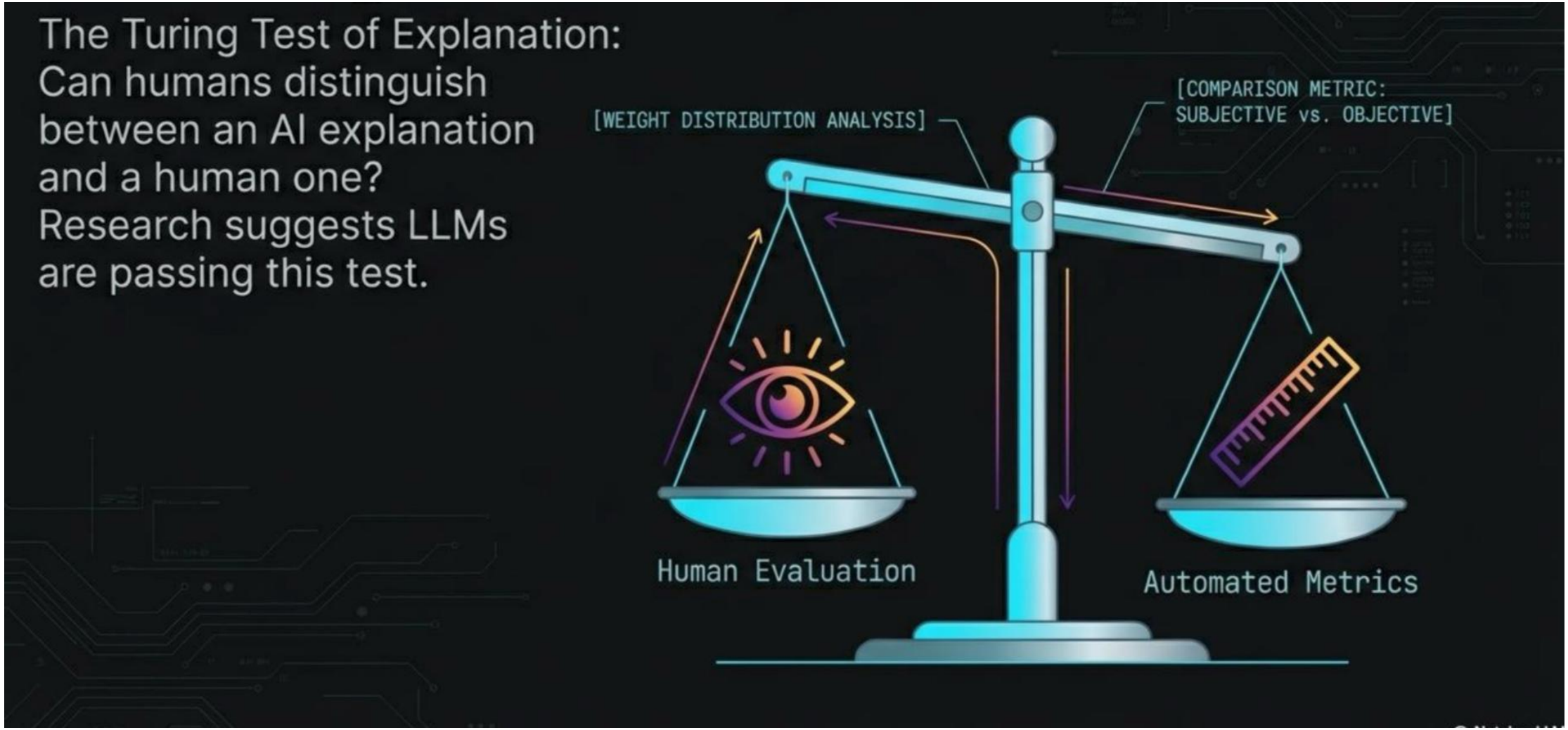
- Explaining how hidden space becomes structure through concept attribution .



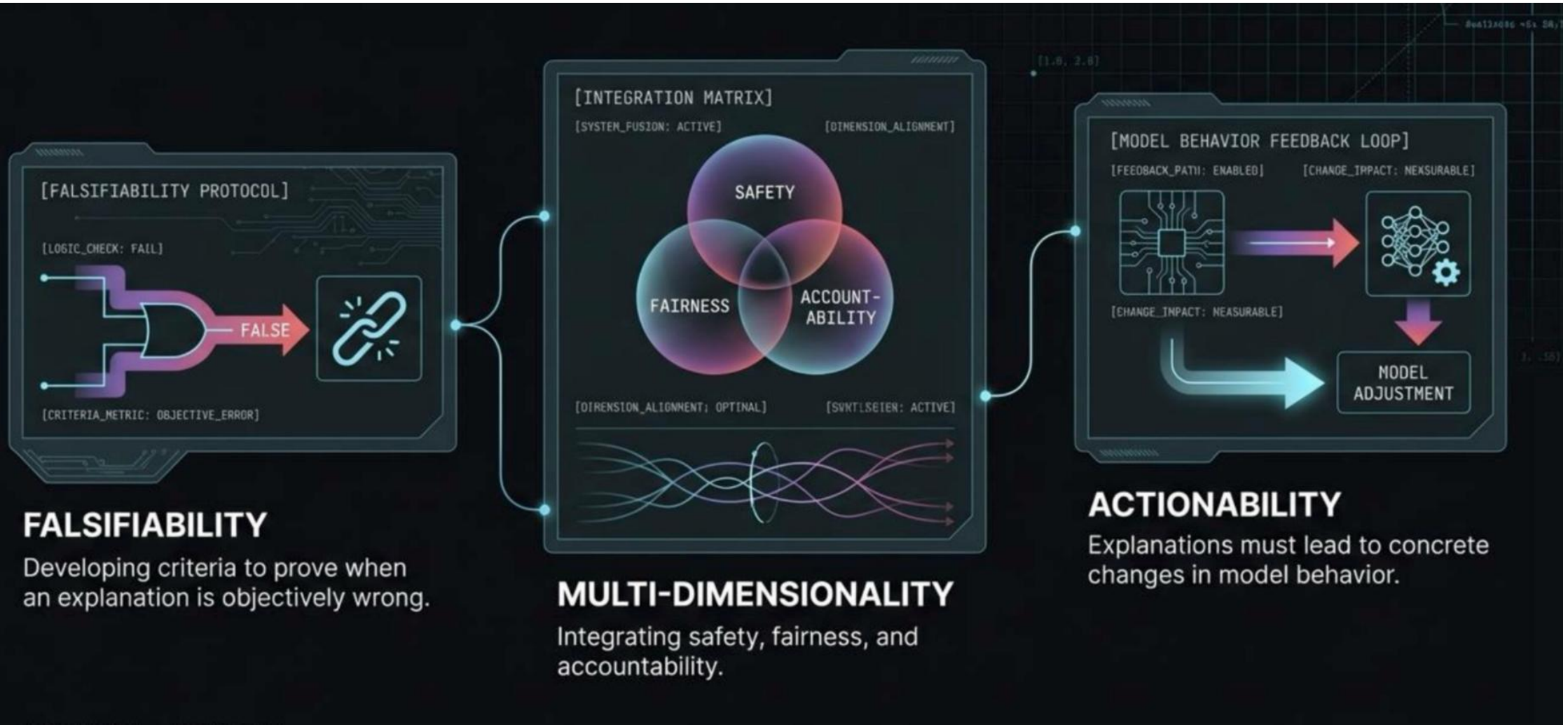
Mechanistic Interpretability



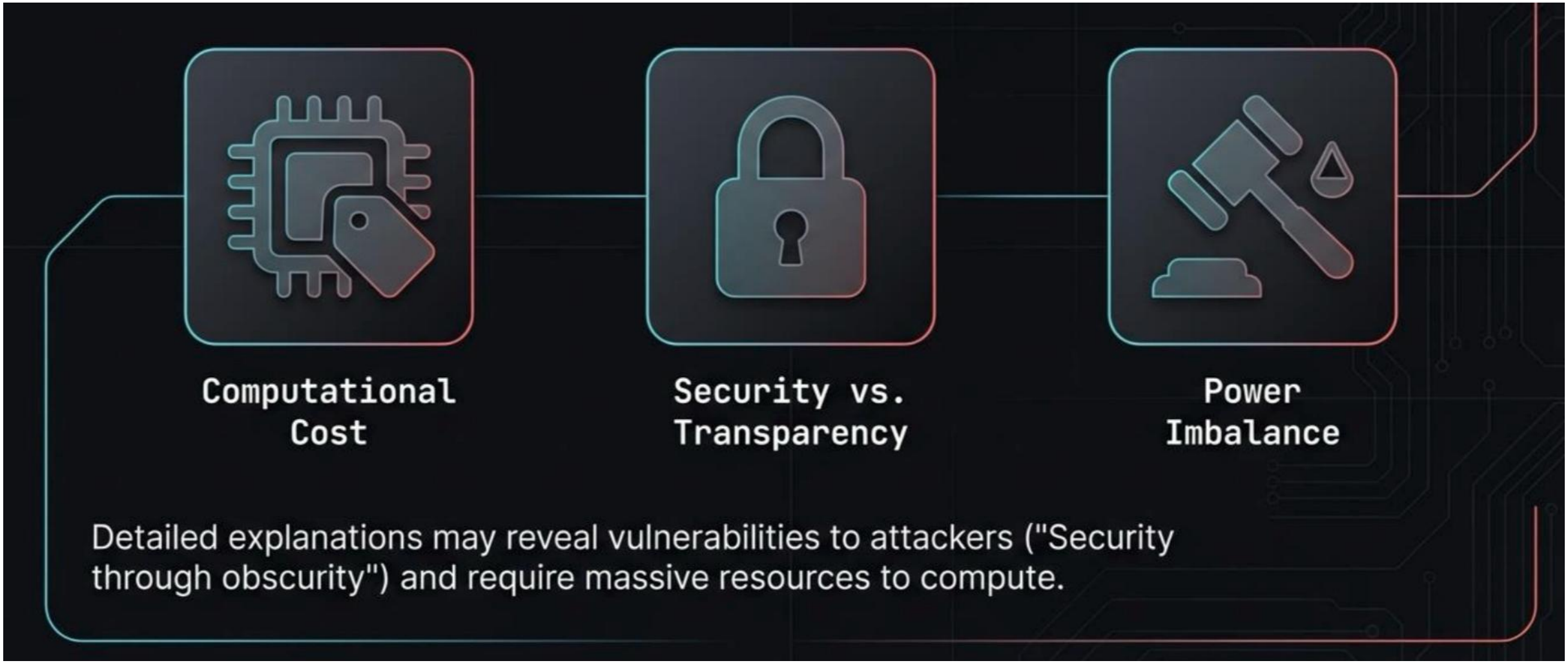
Evaluating Human-like

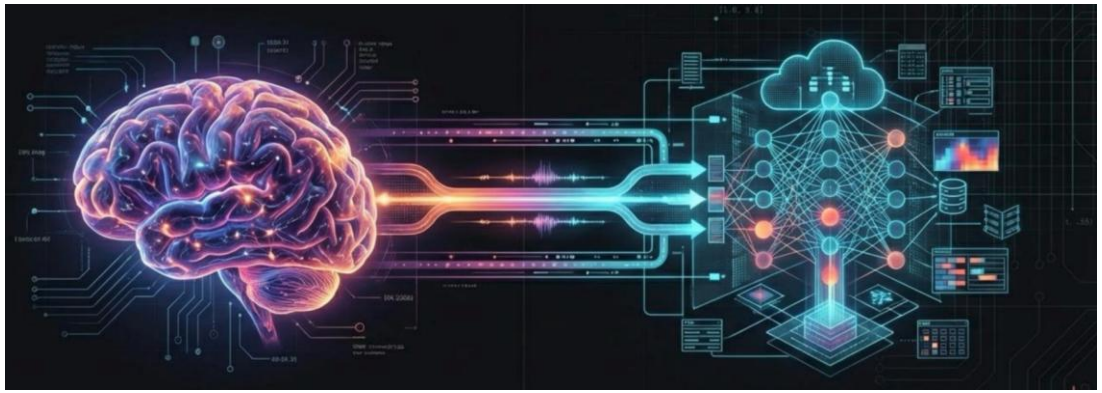


Next?


[FALSIFIABILITY PROTOCOL]
[LOGIC_CHECK: FAIL]
FALSE
[CRITERIA_METRIC: OBJECTIVE_ERROR]
[INTEGRATION MATRIX]
[SYSTEM_FUSION: ACTIVE]
[DIMENSION_ALIGNMENT]
SAFETY
FAIRNESS
ACCOUNT-
ABILITY
[DIMENSION_ALIGNMENT: OPTIMAL]
[SYNTHESIS: ACTIVE]
[MODEL BEHAVIOR FEEDBACK LOOP]
[FEEDBACK_PATH: ENABLED]
[CHANGE_IMPACT: MEASURABLE]
[CHANGE_IMPACT: MEASURABLE]
MODEL
ADJUSTMENT
FALSIFIABILITY
Developing criteria to prove when an explanation is objectively wrong.
MULTI-DIMENSIONALITY
Integrating safety, fairness, and accountability.
ACTIONABILITY
Explanations must lead to concrete changes in model behavior.

Open Challenges and Limitations



HUST

Thanks